Training Day 14

1. Topics Covered and Brief Explanation

1. Introduction to Decision Trees

A Decision Tree is a supervised machine learning algorithm used for classification and regression tasks. It makes predictions by splitting data into smaller groups based on different conditions.

2. Structure of a Decision Tree

Studied the main components of a Decision Tree:

- Root Node: Represents the starting point of the tree.
- Decision Node: Represents a condition or test applied to the data.
- Branch: Represents the outcome of a decision.
- Leaf Node: Represents the final prediction or result.

3. Working Principle of Decision Trees

Learned how a Decision Tree selects features, creates conditions and divides the dataset into smaller groups until it reaches a final prediction.

4. Introduction to K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to divide a dataset into a specified number of groups called clusters.

It groups similar data points together based on their distances from cluster centres (centroids).

5. Working Principle of K-Means

- Select the number of clusters (K).
- Initialize cluster centres.
- Assign each data point to its nearest cluster centre.
- Recalculate the cluster centres.
- Repeat the process until the cluster assignments become stable.

6. Applications

- Decision Trees: Loan approval, customer classification and medical diagnosis support.
- K-Means: Customer segmentation, grouping similar products and organizing data.

2. Tools and Technologies Used

Tool	Purpose
Python	Implementing machine learning algorithms.

Scikit-learn	Creating Decision Tree and K-Means models.
Pandas	Managing datasets.
NumPy	Performing numerical calculations.
Matplotlib	Visualizing data and results.

3. Practical Learning / Code Examples

Example 1: Decision Tree Classification

```
from sklearn.tree import DecisionTreeClassifier
# Sample training data
X = [[1], [2], [3], [4], [5], [6]]
y = ["Low", "Low", "Low", "High", "High", "High"]
model = DecisionTreeClassifier(random_state=0)
model.fit(X, y)
result = model.predict([[5]])
print("Predicted Category:", result[0])
```

Example 2: K-Means Clustering

```
from sklearn.cluster import KMeans
X = [[1, 2], [2, 3], [3, 2],[8, 9], [9, 8], [10, 9]]
model = KMeans(n_clusters=2, random_state=0, n_init=10)
model.fit(X)
print("Cluster Labels:", model.labels_)
```

Explanation:

- DecisionTreeClassifier() creates a Decision Tree model.
- fit() trains the model using sample data.
- KMeans(n_clusters=2) creates two clusters.
- labels_ displays the cluster assigned to each data point.

4. Learning Outcomes

- Understood the structure and working principles of Decision Trees.
- Learned the difference between classification and clustering.
- Gained knowledge of K-Means Clustering and centroids.